Ritu Raj

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TECHNICAL SKILLS

Programming Languages: Golang, Python, Java ,C++
Tech Stack: Redis, Kafka, MySQL, ClickHouse, Grafana, Prometheus, Distributed Systems, Microservices
Others: React, TypeScript, Amazon S3, Spark

SUMMARY

Senior Backend Engineer and early member of Shopee's Anti-Fraud team, with 6+ years building distributed, real-time systems across fraud detection, rule engines, and internal platforms. Delivered multiple 0→1 products end-to-end,from MVP to team-wide adoption,spanning infrastructure (storage migration, cache decoupling), platform (strategy lifecycle, rule engine at 200k+ QPS across 8 countries), and customer-facing products.

WORK EXPERIENCE

Shopee, Singapore — **Anti-Fraud Business Security team**
Senior Software Engineer — *January 2022 - Present*

- **AKV Platform (0→1 Launch):** Built Shopee's anti-fraud strategy lifecycle platform MVP in 3 weeks, enabling real-time fraud projection and strategy evaluation. Led the design of review and audit modules with **custom workflow integration** and developed reliable batch sync jobs for continuous data alignment. Successfully migrated **40% of users** from the legacy platform within the first two quarters.
- **Alarm Notification Centre:** Core contributor to the centralized alerting platform with built-in service health monitoring and health-check capabilities. Introduced false-positive metrics and automated severity downgrade to reduce incident noise. Integrates with **10+ microservices** and processes **50k+ alerts daily**.
- **Audit Service** : Developer and maintainer of a standalone microservice for ingesting team audit logs, built with ClickHouse and Flink. Designed to handle high-throughput workloads, currently supports **200k QPS** with **p99 latency of 200ms**, inspired by Uber's flattened log design.
- **TiDB Storage Migration (Infra focus):** Orchestrated end-to-end migration from TiDB to MySQL across core fraud systems, ensuring zero downtime and full data integrity. Optimized schema and partitioning to cut storage from **0.6TB to 0.3TB/month**, reducing infra cost and boosting query performance.
- **Efficiency Center (MVP → Productized)** : Initiated and led development of a platform that scrapes infrastructure and business metrics to auto-generate sunset recommendations for underperforming fraud strategies. Started as an internal MVP, later adopted by two teams. Saved **700GB+** of cache storage and enabled deprecation of **600+** low-performing strategies, directly reducing infrastructure cost and operational complexity.
- **Technical Leadership:**Drove architecture reviews, code quality governance, and onboarding acceleration across multiple microservices, enabling new engineers to deliver production-ready fraud detection features within 2 months..

Shopee, Singapore — **Anti-Fraud Rule Engine team**
Software Engineer-Early founding engineer (5th member) — *March 2020 - December 2021*

- **Anti-Fraud Rule Engine (Core Contributor):** Core contributor to v2 Shopee's fraud detection backend platform, deployed in **8+ countries** and handling **200k+ QPS**. Led the design of an async microservice processing **30+ real-time events**, boosting detection accuracy and system scalability.
- **Onboard Data Science Team APIs:** Integrated ML-based chargeback REST APIs into the backend engine for **intelligent decisioning**. Extended coverage to **15+ business events**, enabling hybrid detection.

Samsung RnD India — **Market Analysis Team**
Software Engineer — *July 2018 - January 2020*

- **Device Performance Logger:** Developed and maintained code to track sluggish performance and battery drainage for A and M series models on the Samsung Big Data portal.

EDUCATION

National University of Singapore — **Singapore**
Masters in Computer Science — *January 2023 - November 2024*

Birla Institute of Technology — **Mesra,India**
Bachelor of Engineering in Electronics and Communication — *July 2014 - May 2018*